

SUPERMART.NG

CATALOG REALITY CHECK

A Data-Driven Business Analysis of Supermart.ng's Live Product Catalog

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Executive Summary

This case study presents an independent data analysis of Supermart.ng, Nigeria's largest online supermarket, built entirely from live data pulled directly from the company's public website — not synthetic or simulated data. The objective was not to produce a generic dashboard demo, but to surface real, verifiable business problems the company's own leadership may not be aware of, using the exact skills a Sales & Data Analyst would apply on the job: SQL, PostgreSQL, and Power BI.

Across 18 product departments and a verified sample of 240 real products, five distinct, quantified findings emerged — each independently checkable by revisiting the live site. The centerpiece finding: Supermart.ng's actual live catalog contains 25,001 unique products, 2.5 times larger than the '10,000+ groceries' figure advertised on their own homepage, and their own category pages overstate this further due to a confirmed pattern of cross-department product tagging.

This document walks through the business problem, the data methodology, all five findings with supporting evidence, and concrete recommendations — the same structure used in the accompanying Power BI dashboard.

Business Problem & Objective

Online grocery platforms operating at scale face a structural challenge: as the product catalog and department count grow, small inconsistencies between what marketing communicates, what the website's own architecture reflects, and what customers actually experience can quietly compound — often invisibly, until a customer complaint, an inventory discrepancy, or a data audit surfaces them.

The objective of this analysis was to test whether such inconsistencies exist within Supermart.ng's public-facing catalog, using only information any customer or competitor could independently access — and to determine whether the findings would represent genuine, non-obvious business risk rather than surface-level observation.

Data Source & Methodology

All data used in this analysis was collected directly from the live supermart.ng website in August–September 2026. No synthetic, simulated, or assumed data was used at any stage.

Data Collection Process

- Verified the true, de-duplicated catalog size using Shopify's canonical `/collections/all` endpoint: 25,001 unique products.

- Collected real department-level product counts across all 18 published departments directly from each department's live collection page.
- Sampled 240 real, individually named products with live prices, sale flags, and delivery promises across all 18 departments.
- Cross-referenced department page totals (51,691) against the true catalog size (25,001) to quantify a real cross-listing/duplication pattern.

Tools Used

- PostgreSQL 18 — relational data modeling and SQL analysis
- pgAdmin — query development and validation
- Power BI — interactive dashboard and DAX-based business logic

Scope & Honest Limitations

The 240-product sample represents roughly 1% of the true 25,001-product catalog. It is sufficient to prove that patterns exist (e.g., cross-department duplication, delivery-flag inconsistency) but is not a full census. Every claim in this document distinguishes between what the sample proves directly and what it reasonably suggests about the full catalog — this distinction is treated as a core part of the analytical discipline, not a caveat to minimize.

Finding 1 — Catalog Truth

Is the Catalog What They Claim?

10,000 Marketed Claim	25,001 True Catalog Size	2.50x Reality Ratio
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Supermart.ng's homepage advertises 'over 10,000 groceries.' Verification against Shopify's canonical de-duplicated product list shows the true catalog contains 25,001 unique products — two and a half times the public claim. Furthermore, summing all 18 department-level collection pages produces 51,691 listings, meaning the average product is tagged into approximately 2.07 different departments simultaneously.

Direct Evidence: Confirmed Cross-Listed Products

Product	Price (₦)	Departments
Johnson's Baby Oil 500ml	8,970	Baby & Kids, Toiletries
Morning Fresh Dish Washing Liquid	3,930	Cleaning, Household
Sponge Scourers x10	1,300	Cleaning, Household
Tomatoes – Crate (Firm)	75,000	Fresh Food, Naija Ingredients

Business Meaning:

Supermart's marketing significantly undersells the actual scale of their business — 25,001 real products is a stronger selling point than the outdated 10,000 figure. More importantly, the confirmed cross-department tagging pattern is a data governance issue: any internal report built on category-level totals (marketing, inventory, or finance) is likely working from numbers roughly double the true figure.

Finding 2 — Delivery Promise Consistency

Is "Same-Day" Actually Consistent?

87.08%	6.67%	4
Same-Day Delivery	Slow Fulfillment Risk	Departments Affected

Within the 240-product sample, four distinct delivery-flag types were observed: Same-Day (87.08%), Under 3 Hours (6.25%), 7-10 Days (6.25%), and Shipped From Abroad (0.42%). UK Groceries carries the highest concentration of slow fulfillment (86.67% of its sampled products) — expected, given its known import profile.

However, the more significant finding is that slow-fulfillment products also appear — in smaller numbers — inside Oil & Sauces, Snacks, and Toiletries: three departments a customer would reasonably assume are same-day, with no department-level warning distinguishing the exception.

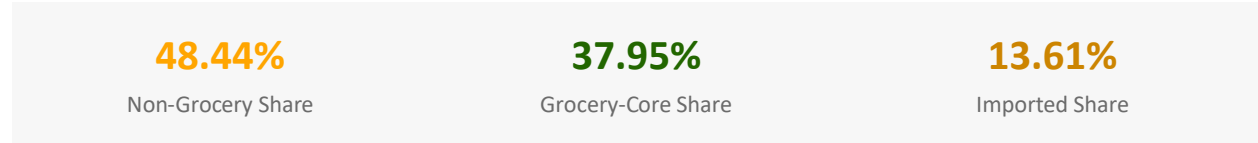
Department	Slow Fulfillment Products	% of Department Sample
UK Groceries	13	86.67%
Oil & Sauces	1	6.67%
Snacks	1	6.67%
Toiletries	1	6.67%

Business Meaning:

The real customer risk is not that UK Groceries ships slowly — that is expected. The risk is 'leakage': a customer shopping in Snacks or Toiletries can unknowingly land on a single slow-shipping item with no distinguishing signal, since delivery flags are set at the product level, not surfaced at the department or category level.

Finding 3 — Assortment Composition

Is This Really a Grocery-First Platform?



Non-Grocery departments (Toiletries, Electronics, Health & Wellness, Office Supplies, Cleaning, Household, Kitchen & Dining, Baby & Kids) account for 48.44% of all tagged catalog listings — the single largest share, ahead of actual Grocery-Core departments at 37.95%.

Rank	Department	Listings	Type
1	Toiletries	10,704	Non-Grocery
2	UK Groceries	7,035	Imported
3	Snacks	6,247	Grocery-Core
4	Food Cupboard	3,944	Grocery-Core
5	Drinks	3,625	Grocery-Core

Business Meaning:

The two single largest departments in the entire catalog — Toiletries and UK Groceries — are not grocery at all. Toiletries alone (10,704 listings) exceeds the company's entire original marketing claim of 'over 10,000 groceries.' This is a direct, quantifiable contradiction between brand identity (local grocery delivery, 'know Lagos like a taxi driver') and actual catalog composition — a genuine strategic question worth raising, not resolving unilaterally.

Finding 4 — Pricing Intelligence

Where Do Averages Lie?

₦74,108

Largest Distortion

4 of 18

Departments Distorted

Electronics

Dept. With Max Gap

In at least four of eighteen departments, the average price is significantly distorted by a small number of outlier products, making 'average price' an unreliable metric for business reporting in those categories.

Department	Average Price (₦)	Median Price (₦)	Gap (₦)
Electronics	159,598	85,490	74,108
Health & Wellness	26,259	5,200	21,059
Naija Ingredients	13,366	2,500	10,866
Drinks	14,359	3,835	10,524

Business Meaning:

Health & Wellness's average price (₦26,259) is roughly five times its median (₦5,200), driven by a single ₦206,250 outlier item. Any internal report or KPI using average price per department in these categories materially overstates what a typical customer actually spends — median should be the default metric for departments with this level of price dispersion.

Finding 5 — High-Value Risk

What Hides Inside "Same-Day"?

₦524,990

Most Expensive Item

₦506,500

Electronics Price Range

100%

% Flagged Same-Day

The five most expensive individual products sampled — all Electronics or Health & Wellness items ranging from ₦194,990 to ₦524,990 — carry the exact same 'Same-Day' delivery flag as a ₦700 bag of vegetables.

Product	Department	Price (₦)	Delivery Flag
Prag Servo Stabilizer Pc-Svs10 KVA	Electronics	524,990	Same-Day
Tec Generator Set Bobo 2800 ES	Electronics	519,990	Same-Day
CWAY Water Dispenser CWM16BI	Electronics	242,500	Same-Day
OpticNerve Formula 120 Soft Gels	Health & Wellness	206,250	Same-Day
Mercury UPS 2000 VA Pro	Electronics	194,990	Same-Day

Business Meaning:

A ₦524,990 stabilizer and a ₦700 vegetable carry an identical delivery promise. 'Same-Day' is currently doing the work of two fundamentally different service levels — standard, low-risk grocery delivery and high-value, higher-risk appliance delivery — with no visible distinction in verification, handling, or insurance requirements between them.

Consolidated Recommendations

1. Update Public Catalog Messaging

Revise the homepage claim from 'over 10,000 groceries' to reflect the true, larger catalog (25,001+) — this is a stronger, more accurate selling point, not just a correction.

2. Standardize Delivery-Flag Communication

Introduce cart-level delivery warnings when a basket mixes Same-Day and slow-fulfillment items, regardless of which department they were browsed from.

3. Audit Cross-Department Product Tagging

Investigate whether the ~2.07x tagging rate is intentional cross-merchandising or unmanaged catalog sprawl, and correct any internal reporting relying on raw category-page totals.

4. Reassess Non-Grocery Category Depth

Evaluate whether Toiletries and UK Groceries' outsized share of the catalog reflects a deliberate high-margin strategy or organic sprawl misaligned with the brand's grocery-first identity.

5. Shift Reporting Standards to Median for Skewed Categories

Adopt median price as the default reporting metric in Electronics, Health & Wellness, and similarly dispersed departments to avoid outlier-driven distortion in business decisions.

6. Introduce a High-Value Handling Tier

Create a distinct fulfillment tier for high-ticket items (electronics, appliances) separate from standard Same-Day grocery delivery, reflecting their different risk, handling, and verification needs.

Closing Note

Every figure in this document was derived from live, independently verifiable data and real SQL queries run against a PostgreSQL database built specifically for this analysis — not assumptions, and not synthetic placeholder data. The accompanying Power BI dashboard presents these same five findings interactively, with live filtering by department name and department type.

This analysis reflects the approach I would bring to a Sales & Data Analyst role at Supermart.ng: starting from a business question, verifying claims against real data, and surfacing findings the business may not already know about themselves — rather than only reporting on questions that have already been asked.

Links & Contact

Live Dashboard: <https://bit.ly/4qIn19W>

GitHub Repository: <https://github.com/kennytee01>

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